

Input variables (inputs)

- 1.connection strength
2. response time
3. energy consumption

Trapezoid membership functions for the input variables can be defined as

$$\mu_{low}(CS) = \begin{cases} 1 & \text{if } CS \leq 20, \\ \frac{40 - CS}{40 - 20} & \text{if } 20 < CS \leq 40, \\ 0 & \text{if } CS > 40. \end{cases}$$

$$\mu_{med}(CS) = \begin{cases} 0 & \text{if } CS \leq 20, \\ \frac{CS - 20}{40 - 20} & \text{if } 20 < CS \leq 40, \\ 1 & \text{if } 40 < CS \leq 60, \\ \frac{80 - CS}{80 - 60} & \text{if } 60 < CS < 80, \\ 0 & \text{if } CS \geq 80. \end{cases}$$

$$\mu_{high}(CS) = \begin{cases} 0 & \text{if } CS \leq 60, \\ \frac{CS - 60}{80 - 60} & \text{if } 60 < CS \leq 80, \\ 1 & \text{if } CS > 80. \end{cases}$$

$$\mu_{low}(RT) = \begin{cases} 1 & \text{if } RT \leq 30, \\ \frac{50 - RT}{50 - 30} & \text{if } 30 < RT \leq 50, \\ 0 & \text{if } RT > 50. \end{cases}$$

$$\mu_{med}(RT) = \begin{cases} 0 & \text{if } RT \leq 30, \\ \frac{RT - 30}{50 - 30} & \text{if } 30 < RT \leq 50, \\ 1 & \text{if } 50 < RT \leq 70, \\ \frac{90 - RT}{90 - 70} & \text{if } 70 < RT < 90, \\ 0 & \text{if } RT \geq 90. \end{cases}$$

$$\mu_{high}(RT) = \begin{cases} 0 & \text{if } RT \leq 70, \\ \frac{RT - 70}{90 - 70} & \text{if } 70 < RT \leq 90, \\ 1 & \text{if } RT > 90. \end{cases}$$

$$\mu_{low}(EC) = \begin{cases} 1 & \text{if } EC \leq 10, \\ \frac{30 - EC}{30 - 10} & \text{if } 10 < EC \leq 30, \\ 0 & \text{if } EC > 30. \end{cases}$$

$$\mu_{med}(EC) = \begin{cases} 0 & \text{if } EC \leq 10, \\ \frac{EC - 10}{30 - 10} & \text{if } 10 < EC \leq 30, \\ 1 & \text{if } 30 < EC \leq 50, \\ \frac{70 - EC}{70 - 50} & \text{if } 50 < EC < 70, \\ 0 & \text{if } EC \geq 70. \end{cases}$$

$$\mu_{high}(EC) = \begin{cases} 0 & \text{if } EC \leq 50, \\ \frac{EC - 50}{70 - 50} & \text{if } 50 < EC \leq 70, \\ 1 & \text{if } EC > 70. \end{cases}$$

Output variable (output)

security risk level

Membership functions for the output variable, which are of triangular type, are determined as

$$\mu_{vl}(SR) = \begin{cases} 1 & \text{if } SR \leq 0, \\ \frac{25 - SR}{25} & \text{if } 0 < SR \leq 25, \\ 0 & \text{if } 25 < SR. \end{cases}$$

$$\mu_{low}(SR) = \begin{cases} 0 & \text{if } SR \leq 0, \\ \frac{SR}{25} & \text{if } 0 < SR \leq 25, \\ \frac{50 - SR}{25} & \text{if } 25 < SR \leq 50, \\ 0 & \text{if } 50 < SR. \end{cases}$$

$$\mu_{med}(SR) = \begin{cases} 0 & \text{if } SR \leq 0, \\ \frac{SR - 25}{25} & \text{if } 25 < SR \leq 50, \\ \frac{75 - SR}{25} & \text{if } 50 < SR \leq 75, \\ 0 & \text{if } 75 < SR. \end{cases}$$

$$\mu_{high}(SR) = \begin{cases} 0 & \text{if } SR \leq 50, \\ \frac{SR - 50}{25} & \text{if } 50 < SR \leq 75, \\ \frac{100 - SR}{25} & \text{if } 75 < SR \leq 100, \\ 0 & \text{if } 100 < SR. \end{cases}$$

$$\mu_{vh}(SR) = \begin{cases} 0 & \text{if } SR \leq 75, \\ \frac{SR - 75}{25} & \text{if } 75 < SR \leq 100, \\ 1 & \text{if } 100 < SR. \end{cases}$$